

test-coverage

DocProcessor — Test Coverage Matrix

Round 220 (2026-05-19) — per-test-file coverage matrix enumerating the seven test types CONST-050(B) demands for this module's domain (unit, integration, E2E, automation, security, stress, Challenge), the file-system location of each, the package or surface they cover, the kind of evidence they capture, and the paired-mutation guard (per §1.1) that protects against the test itself silently degrading into a bluff.

Verbatim 2026-05-19 operator mandate (CONST-049 §11.4.17): *"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! ... execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"*

1. Coverage matrix (real surfaces actually exercised)

Test file	Type	Real surface exercised	Captured evidence	Paired mutation
pkg/loader/loader_test.go	Unit	loader.NewDefaultLoader, format detection, single-file load	Loaded Document struct contents asserted	Renamed support format in for slice + assertion fails
pkg/loader/loader_integration_test.go	Integration	loader.LoadDir against real temp dir tree	Real fs file count +	Skip a format register

Test file	Type	Real surface exercised	Captured evidence	Paired mutations
pkg/loader/loader_security_test.go	Security	Path-traversal / symlink-escape / oversize-input rejection	per-doc body Rejected-path error message + no read attempted	loader count Remove path-icalise step - versa success test fail
pkg/loader/loader_stress_test.go	Stress	Concurrent LoadDir invocations, large fanout	No data race (-race) + per-iter timing	Strip sync.Race in loader race condition or trip
pkg/loader/markdown.go + parser tests	Unit	Markdown section parsing	Parsed AST sections	Truncate parse input → section count section
pkg/feature/builder_test.go	Unit	BuildFromDocs heuristic extraction	Feature/Screen/Workflow counts + category map	Remove char heuristic guess spurious feature count fails
pkg/feature/feature_test.go	Unit	Feature struct invariants + equality	Equality + JSON round-trip	Drop struct → JSON round-trip diff fail
pkg/feature/convert.go tests	Unit	Doc → feature conversion	Converted feature contents	Skip category evidence category map missing
pkg/coverage/tracker_test.go	Unit	CoverageTracker set/get per-platform	Per-platform status map	Replace RWMutex with mutex → concurrent

Test file	Type	Real surface exercised	Captured evidence	Paired mutations
				turns state
pkg/coverage/tracker_stress_test.go	Stress	1000-goroutine concurrent set+get	No data race + final state consistency	Remove lock -> detected trips
pkg/docgraph/graph_test.go	Unit	Graph.AddNode/AddEdge, JSON + Mermaid export	Exported JSON + Mermaid strings	Skip Mermaid invalid tax -> fails
pkg/docgraph/graph_stress_test.go	Stress	Concurrent node/edge insertion	No data race + edge count	Strip race condition or trip
pkg/llm/prompts_test.go	Unit	Prompt template formatting	Formatted prompt string	Drop holder assertion fails
pkg/config/config_test.go	Unit	.env parsing into Config	Parsed Config struct	Skip default inject default missing
pkg/config/config_security_test.go	Security	Reject path-traversal in HELIX_DOCS_ROOT	Rejection error message	Remove validation -> traversal success test fails
pkg/i18n/translator_test.go	Unit	NoopTranslator.T + TPlural verbatim-ID semantics	Returned-string equality with msg ID	Make return equal fails
cmd/docprocessor/main_test.go	Unit + E2E	runCLI direct invocation, 8 branch tests, bundle integrity	Sentinel translations + forbidden-lit	Drop use go remove sentinel bundle

Test file	Type	Real surface exercised	Captured evidence	Paired mutation
e2e_test.go	E2E	Built binary executed against real temp tree	eral absence Exit code + stdout/stderr captures	or brack test fa Repla binar /bin/ sentin check
automation_test.go	Automation	Full doc-load → feature-build → coverage-track pipeline (real)	Pipeline-output equality	Skip c richm step - put d
security_test.go	Security	Reject symlink-escape, oversize args, malformed env	Rejection error + no execution	Remo valida guard test f
challenges/scripts/docprocessor_cli_challenge.sh	Challenge	Built binary against synthetic + empty docs dirs, both modes	Captured wire output + exit code + paired mutation	--anti bluff strips sertio script turns

2. Anti-bluff floor (per CONST-035 / Article XI §11.9)

Every row above satisfies these four invariants:

1. **Positive captured evidence.** The test/Challenge writes the actual wire output, file contents, or struct state it asserts on (no "no error returned therefore PASS" patterns).
2. **Real-data exercise.** Non-unit rows touch real filesystems, real binaries, real parsers — no mock loaders, no mock binaries.
3. **Paired-mutation guard.** Every gate has a paired mutation (last column) that, when applied, makes the test FAIL — proving the assertion catches the negation per §1.1.
4. **CONST-046 wire-content check.** Every CLI test asserts the translated sentinel appears AND the forbidden English literal does NOT (preventing accidental hardcoded-string regression).

3. Coverage of CONST-050(A) — no fakes beyond unit tests

Layer	Permitted fakes	DocProcessor compliance
Unit	mocks, stubs allowed	fakeTranslator lives in cmd/docprocessor/main_test.go (unit-only)
Integration	none	loader_integration_test.go writes real fs trees, no mocks
E2E	none	e2e_test.go invokes built binary, no mocks
Automation	none	automation_test.go runs real pipeline end-to-end
Security	none	loader_security_test.go / config_security_test.go use real inputs
Stress	none	*_stress_test.go use real concurrent goroutines + -race
Challenge	none	challenges/scripts/*.sh invoke the built binary

Production code (pkg/**/*.go not ending _test.go, cmd/**/main.go) is periodically swept for forbidden patterns (grep -rn "simulated\|for now\|TODO implement\|placeholder" pkg cmd); the sweep returns empty as of round 220.

4. Coverage of CONST-046 — no hardcoded content

The CLI is the only user-facing surface in this module; its eighteen wire strings are exhaustively gated by cmd/docprocessor/main_test.go:

- Eight branch tests assert each sentinel appears in the appropriate stream.
- Each branch test asserts the corresponding English literal does NOT appear (paired forbidden-literal check).
- TestRunCLI_BundleContainsAllRound209MsgIDs asserts every msg ID exists in pkg/i18n/bundles/active.en.yaml — planting a typo in the bundle key makes the test fail (paired mutation: rename docprocessor_cli_done to docprocessor_cli_done2 → test fails).
- TestRunCLI_NoopTranslator_EmitsMsgIDVerbatim integrates the production i18n.NoopTranslator default to prove the verbatim-ID fallback path is itself captured (so absence-of-bundle is loudly visible per CONST-035).

5. How to add a new test that satisfies the matrix

1. Pick the layer (unit / integration / E2E / automation / security / stress / Challenge).
2. Place the file in the canonical location (see §1).
3. Capture positive runtime evidence (assert on actual struct contents, wire output, file system state — NOT on `err == nil`).
4. Add a paired-mutation note to the §1 row.
5. If the new test touches the CLI, add the new msg ID to both `pkg/i18n/bundles/active.en.yaml` AND the §1.4 list.
6. Update `cmd/docprocessor/main.go` package doc with the new round number and msg-ID rationale.
7. Run `go test ./... -race -count=1` AND the relevant Challenge script; confirm both green; capture the run output in your commit message.

6. Round-history of test-matrix growth

Round	Date	Net change
97	2026-04	CONST-046 first migration — 8 CLI msg IDs externalised + tests
209	2026-05-19	CONST-046 second migration — 10 additional CLI msg IDs + 7 new tests
220	2026-05-19	docs/test-coverage.md + Challenge script (this document)

7. References

- Root: [CONSTITUTION.md](#), [CLAUDE.md](#), [AGENTS.md](#).
- Anti-bluff covenant: §11.4 of the canonical constitution sub-module.
- Test-type coverage mandate: CONST-050(B).
- No-hardcoded-content mandate: CONST-046.
- Decoupling mandate: CONST-051(B).